THE FIRST COHOMOLOGY OF AN O-MODULE AGREES WITH THE FIRST COHOMOLOGY OF ITS UNDERLYING ABELIAN SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author writes 'The converse is a little more fun' and omits the double-composition check; both are reproduced as written.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; the Xy-pic xy environment is required because the counted proof displays two maps of short exact sequences. No macro was invented or redefined.

Editorial note. The same-chapter reference is located at its \label line inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- The bijection between Ext^1 and H^1 of the structure sheaf. Same chapter, Lemma labelled lemma-h1-extensions (source line 372, Stacks tag 03F1); invoked at proof line 442. Supplies the translation of both sides of the asserted isomorphism into groups of extensions, which is the first step of the proof; the two comparison constructions are the counted content.
- Tensoring with a flat module preserves exactness. Modules on Sites, Lemma labelled sites-modules-lemma-flat-tor-zero (canonical Stacks tag 03EX); invoked at proof lines 480-481. Supplies exactness of the sequence obtained from an extension of Z by F_ab after tensoring with O; the pushout construction on top of it is retained in the counted span.

Source reference. Original lemma 03F1: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 372–372. Same-chapter pointer invoked at proof line 442 to translate both H^1 groups into extension groups; the two comparison constructions that form the counted argument are retained verbatim in the delivered body.

Source reference. Original lemma 03EX: sources/stacks-chow/tags.txt, source lines 4472–4472. Cross-chapter standard prerequisite invoked once at proof lines 480-481 for exactness after tensoring with O; the pushout construction on top of it is retained in the counted span.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules on $\mathcal{C}$. Let $\mathcal{F}_{ab}$ denote the underlying sheaf of abelian groups. Then there is a functorial isomorphism*

$$H^1(\mathcal{C}, \mathcal{F}_{ab}) = H^1(\mathcal{C}, \mathcal{F})$$

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where the left hand side is cohomology computed in $Ab(\mathcal{C})$ and the right hand side is cohomology computed in $Mod(\mathcal{O})$.

Proof. Let $\underline{\mathbf{Z}}$ denote the constant sheaf $\mathbf{Z}$. As $Ab(\mathcal{C}) = Mod(\underline{\mathbf{Z}})$ we may apply Lemma 03F1 twice, and it follows that we have to show

$$\mathrm{Ext}^1_{Mod(\mathcal{O})}(\mathcal{O}, \mathcal{F}) = \mathrm{Ext}^1_{Mod(\underline{\mathbf{Z}})}(\underline{\mathbf{Z}}, \mathcal{F}_{ab}).$$

Suppose that $0 \to \mathcal{F} \to \mathcal{E} \to \mathcal{O} \to 0$ is an extension in $Mod(\mathcal{O})$. Then we can use the obvious map of abelian sheaves $1 : \underline{\mathbf{Z}} \to \mathcal{O}$ and pullback to obtain an extension $\mathcal{E}_{ab}$, like so:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F}_{ab} & \longrightarrow & \mathcal{E}_{ab} & \longrightarrow & \underline{\mathbf{Z}} & \longrightarrow & 0 \\ & & \| & & \downarrow & & \downarrow{\scriptstyle 1} & & \\ 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{E} & \longrightarrow & \mathcal{O} & \longrightarrow & 0 \end{array}$$

The converse is a little more fun. Suppose that $0 \to \mathcal{F}_{ab} \to \mathcal{E}_{ab} \to \underline{\mathbf{Z}} \to 0$ is an extension in $Mod(\underline{\mathbf{Z}})$. Since $\underline{\mathbf{Z}}$ is a flat $\underline{\mathbf{Z}}$-module we see that the sequence

$$0 \to \mathcal{F}_{ab} \otimes_{\underline{\mathbf{Z}}} \mathcal{O} \to \mathcal{E}_{ab} \otimes_{\underline{\mathbf{Z}}} \mathcal{O} \to \underline{\mathbf{Z}} \otimes_{\underline{\mathbf{Z}}} \mathcal{O} \to 0$$

is exact, see Modules on Sites, Lemma 03EX. Of course $\underline{\mathbf{Z}} \otimes_{\underline{\mathbf{Z}}} \mathcal{O} = \mathcal{O}$. Hence we can form the pushout via the ($\mathcal{O}$-linear) multiplication map $\mu : \mathcal{F} \otimes_{\underline{\mathbf{Z}}} \mathcal{O} \to \mathcal{F}$ to get an extension of $\mathcal{O}$ by $\mathcal{F}$, like this

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F}_{ab} \otimes_{\underline{\mathbf{Z}}} \mathcal{O} & \longrightarrow & \mathcal{E}_{ab} \otimes_{\underline{\mathbf{Z}}} \mathcal{O} & \longrightarrow & \mathcal{O} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mu} & & \downarrow & & \| & & \\ 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{E} & \longrightarrow & \mathcal{O} & \longrightarrow & 0 \end{array}$$

which is the desired extension. We omit the verification that these constructions are mutually inverse. □